AI SECURITY & INTEGRITY

Are You Getting What You Pay For? Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

THE PROBLEM

The Model Substitution Problem: Economic Incentives Drive Covert Swa

The Trust Gap in Commercial APIs

{{icon:finance/money-bill-wave}}

Providers charge for specific models (e.g., 70B), but face strong economic incentives to covertly substitute cheaper alternatives.

Four Types of Covert Substitution (H0: Genuine vs. H1: Substituted)

1. Smaller Model

Serving a smaller parameter count model. e.g., Llama-3-8B instead of 70B.

2. Quantized Variant

Serving lower precision weights to save VRAM. e.g., FP8/INT8 instead of FP16.

3. Fine-tuned Version

Serving a modified or updated version without notifying the customer.

4. Different Family

Serving an entirely different architecture from a cheaper provider.

EMPIRICAL EVIDENCE

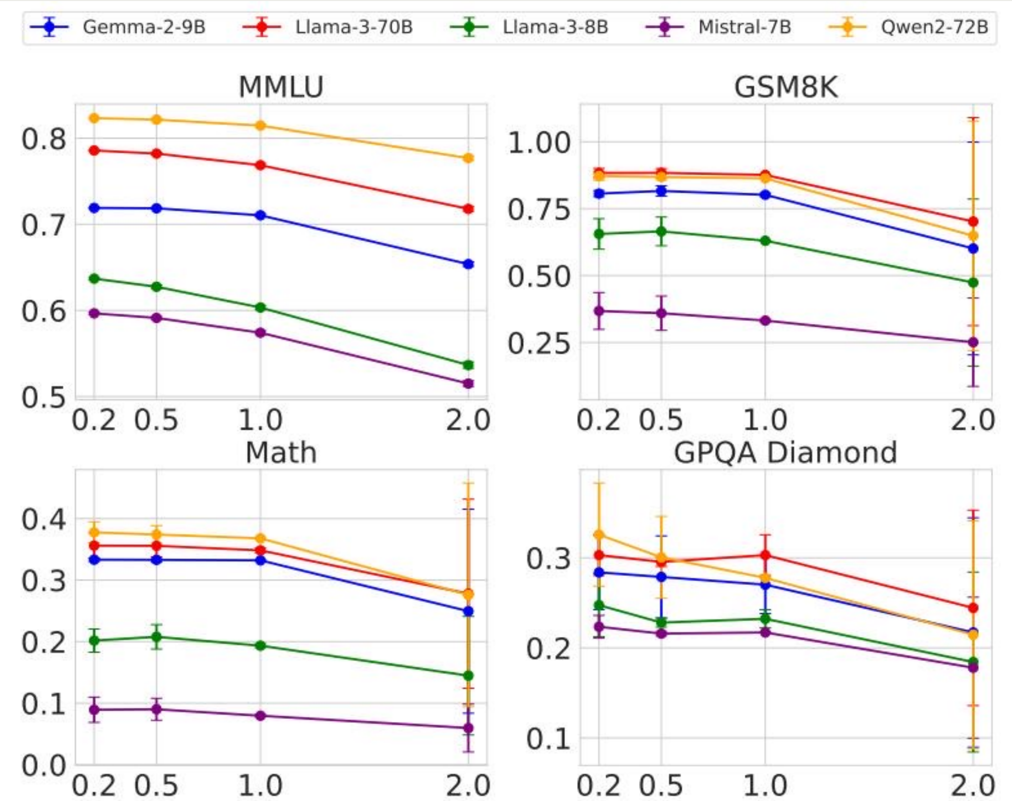
FP8 Quantization Preserves Most Accuracy — Making Substitution Hard

The drop in performance from FP16 to FP8 is minimal, often falling within the statistical error bars, completely masking the substitution.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct (Original)	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct (Original)	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Qwen2-72B-Instruct (Original)	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3 (Original)	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

EMPIRICAL EVIDENCE

Higher Temperature Amplifies Performance Gaps but Increases Noise



Temperature Scaling Fails

- High temperatures (e.g., T=2.0) uniformly degrade performance across all models.
- As temperature increases, the statistical gap between the original model and the substitute narrows or is consumed by statistical noise.
- This limits the auditor's ability to use temperature scaling as an effective hypothesis testing signal.

METADATA FAILURE

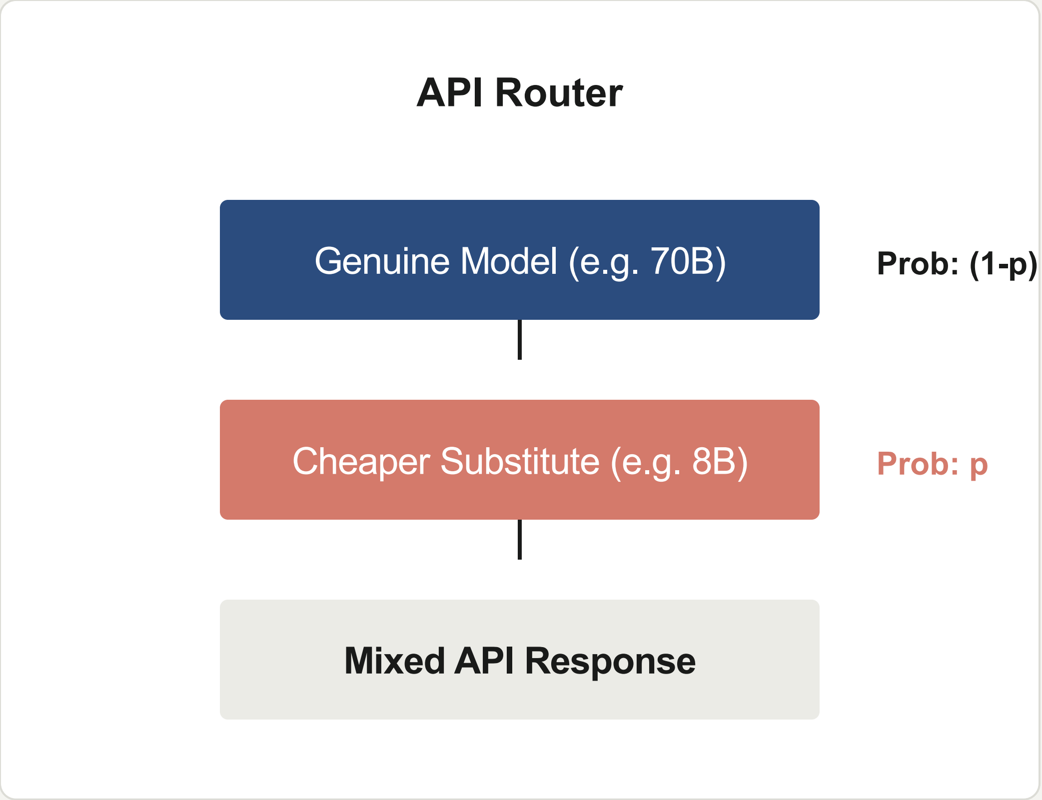
Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

System Variation Breaks the Math

- Log probability comparisons fail because of inherent inference nondeterminism in production.
- The same model (e.g., Llama-3-8B) produces varying log-probs depending on the software framework (transformers 4.40 vs 4.50, vLLM) and GPU hardware (A100 vs H100).

Randomized Routing Makes Statistical Detection Query-Prohibitive



Adversarial Countermeasures

- **Randomized Substitution**
  Providers can mix substitute responses with probability p.
  As $p \to 0$, the mixed distribution converges to the genuine distribution.
  Exponentially more queries are required for reliable detection.
- **Benchmark Evasion**
  Detect known benchmark prompts and route them to the genuine model while serving substitutes for regular traffic.

SUMMARY OF FAILURES

Software-Only Auditing Methods Are Fundamentally Unreliable

The three main categories of software-only auditing all experience critical, unresolvable failure modes in production.

Statistical Tests
(e.g., MMD)

Failure Mode:

Require impractically large query budgets and fail to detect subtle substitutions like FP8 quantization where the output distributions are nearly identical.

Log-Probability
(Token Fingerprinting)

Failure Mode:

Completely defeated by legitimate production-level nondeterminism. Varying frameworks and hardware naturally shift token probs, causing false positives.

Adversarial Evaders
(Routing & Caching)

Failure Mode:

Randomized routing and benchmark evasion explicitly bypass software detection boundaries, making the API practically a moving target.

THE SOLUTION

Trusted Execution Environments Provide Provable Cryptographic Integr

Why Hardware Attestation Wins

1. Provable Attestation

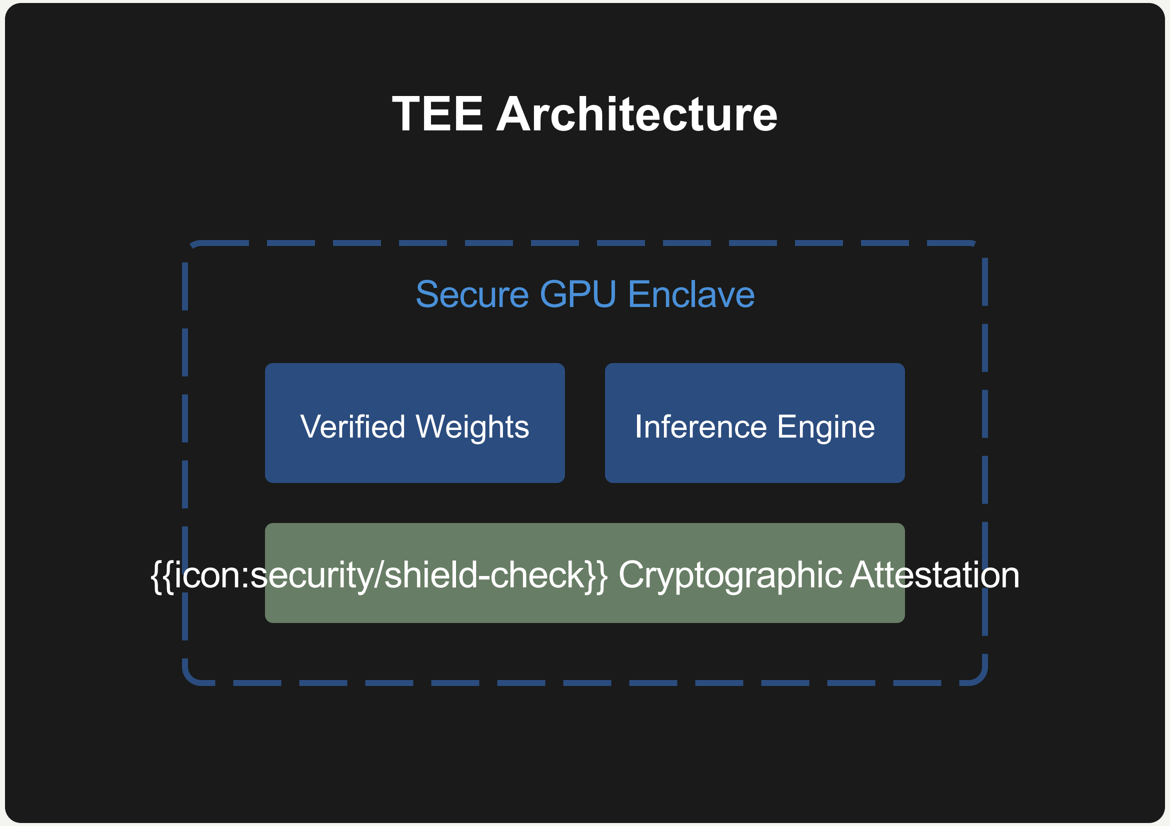
TEEs mathematically guarantee specific weights are loaded.

2. Hardware Security

Enclaves secure data during execution (e.g., NVIDIA H100).

3. Efficient & Robust

Unlike ZKPs, TEEs have modest overhead, deployable today.



CONCLUSION

Key Takeaways: Close the Verification Gap with Hardware Attestation

- **Model substitution is economically motivated and practically undetectable via software.**
- **FP8 quantization barely affects benchmarks, serving as the perfect covert swap.**
- **Inference nondeterminism naturally defeats log-probability fingerprinting.**
- **Randomized substitution explicitly bypasses statistical text-based tests.**

The Actionable Path Forward

TEEs are the only currently viable solution for verifiable, trustworthy LLM APIs.